

Green tea: nature's defense against malignancies.

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The current practice of introducing phytochemicals to support the immune system or fight against diseases is based on centuries old traditions. Nutritional support is a recent advancement in the domain of diet-based therapies; green tea and its constituents are one of the important components of these strategies to prevent and cure various malignancies. The anti-carcinogenic and anti-mutagenic activities of green tea were highlighted some years ago suggesting that it could reduce the prevalence of cancer and even provide protection. The pharmacological actions of green tea are mainly attributed to polyphenols that includes epigallocatechin-3-gallate (EGCG), epicatechin, epicatechin-3-gallate, epigallocatechin. Green tea and its components effectively mitigate cellular damage arising due to oxidative stress. Green tea is supposed to enhance humoral and cell-mediated immunity, decreasing the risk of certain cancers, and may have certain advantage in treating inflammatory disorders. Much of the cancer chemopreventive properties of green tea are mediated by EGCG that induces apoptosis and promotes cell growth arrest, by altering the expression of cell cycle regulatory proteins, activating killer caspases, and suppressing nuclear factor kappa-B activation. Besides, it regulates and promotes IL-23 dependent DNA repair and stimulates cytotoxic T cells activities in a tumor microenvironment. It also blocks carcinogenesis by modulating the signal transduction pathways involved in cell proliferation, transformation, inflammation and metastasis. The review is intended to highlight the chemistry of green tea, its antioxidant potential, its immunopotentiating properties and mode of action against various cancer cell lines that showed its potential as a chemopreventive agent against colon, skin, lung, prostate, and breast cancer.